APGENCO	ITEM: ERECTION OF NEW VANES / REPAIR OF DAMAGED INLET GUIDE VANES OF PARALLEL ESP'S SUB-SYSTEM:PARALLEL ESPs GUIDE VANES	**STANDARD FIELD QUALITY PLAN**	QP NO.: APG/QC-3/SFQP:300 REV. NO.:00 DATE: 06.05. 2024 PAGE: 1…. OF….3	Reviewed By: EE/QC3 REVIEWED By EE QC3 at 9:34 am, May 10, 2024 Approved By: SE/QC APPROVED By EE/QC at 3:46 pm, May 10, 2024

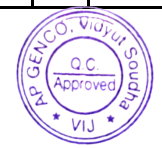
SL. NO	ACTIVITY ANDOPERATION	CHARACTERISTICS / INSTRU-MENTS	CLASS OF CHECK #	TYPE OF CHECK	QUANTUM OF CHECK	REFERENCE DOCUMENT	ACCEPTANCE NORMS	FORMAT OF RECORD	D*	REMARKS
1.	2.	3.	4.	5.	6.	7.	8.	9.		10.
1	ERECTION OF NEW VANES IN PARALLEL ESP'S AT INLET DUCT 1. TRANSPORTATION OF TOOLS AND TACKLES AND MATERIALS FROM STORES. 2. TRANSPORTATION AND SHIFTING OF GUIDE VANES FROM ' 0' MTRS TO 24 MTRS ELEVATION BY CHAIN PULLEY BLOCK/ HOIST 3. CUTTING OF F G DUCT ON TOP FOR INSERTING THE GUIDE VANE. 4. ERECTION OF NEW INLET GUIDE VANES IN IT ORIGINAL POSITION AND SEAM WELDING WITH DUCT WALLS. 5. FOR REPAIRS TO INLET GUIDE VANE :- IN CASE OF ERECTION OF NEW GUIDE VANES ARE NOT POSSIBLE THE FIRM SHOULD CARRY REPLACE THE DAMAGED PORTION OF VANE DULY FABRICATED IN TO ORIGINAL SHAPE. 6. REMOVAL OF SUPPORTS AFTER COMPLICATION OF WORK. 7. TRANSPORTING BACK ALL TOOLS AND TACKLES TO STORES. 8. CLEANING THE AREA AFTER COMPLETION OF WORK	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STAND-ARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)
2	DISMANTLING OF DAMAGED GUIDE VANES IN PARALLEL ESP'S AT INLET DUCT 1. TRANSPORTATION OF TOOLS AND TACKLES AND MATERIALS FROM STORES 2. ARRESTING OF GUIDE VANES WITH DUCT WALL AND CUTTING OF DAMAGED GUIDE VANES IN TO PIECES. 3. REMOVAL OF CUT GUIDE VANE PIECES AND TRANSPORTING THE PIECES FROM 24 MTRS ELEVATION TO 'O' MTRS ELEVATION BY CHAIN PULLEY BLOCK. 4. DEVOLUTION OF SCRAP TO STORES. 5. TRANSPORTING BACK ALL TOOLS AND TACKLES AND MATERIALS TO STORES. 6. CLEANING THE AREA AFTER COMPLETION OF WORK.	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STAND-ARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)

APGENCO, Vidyut Soudha, Q.C. Approved, VIJ

3	REPAIRS TO DUCT WALLS AND SEAL PLATES OF EXPANSION JOINTS IN BOILER I&II FG DUCTS 1. TRANSPORTING OF M.S PLATE 3.15 / 5MM THICK AND EXPANSION JOINT FROM STORES TO SITE. 2. INSPECTION OF DUCT EXPANSION JOINTS, SEAL PLATES, MARKING OF THE DAMAGED OR ERODED PORTIONS. 3. CUTTING OF THE DAMAGED PORTIONS OF THE DUCT WALLS, LINER PLATES AND SEAL PLATES AND TAKING OUT SIDE THE DUCTS AND KEEPING IN A SPECIFIED PLACE 4. CUTTING OF THE NEW M.S PLATE (SIZE 3.15/5MM THICK) INTO REQUIRED SIZES TRANSPORTING THEM INSIDE THE DUCTS LOCATION AND FIXING THE PLATES DULY FULL WELDING.	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)
4	PROVIDING OF PLATFORM AND SCAFFOLDING INSIDE THE DUCT. 1. TRANSPORTATION OF SCAFFOLDING PIPES, CLAMPS AND WOODEN PLANKS INSIDE THE DUCT FROM STORES 2. ARRANGING SCAFFOLDING PIPES AND WOODEN PLANKS FOR CUTTING AND ERECTION OF THE INLET GUIDE VANES. 3. DISMANTLING THE SCAFFOLDING AND PLATFORM AFTER COMPLETION OF THE WORK 4. TRANSPORTING BACK ALL THE SCAFFOLDING PIPES, CLAMPS AND WOODEN PLANKS TO STORES. 5. CLEANING THE AREA AFTER COMPLETION OF WORK	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)

5	REPAIR OR REPLACEMENT OF DAMAGED PORTION IN HONEY COMB AREA AND BRACINGS & ANGLES. 1. TRANSPORTATION OF TOOLS AND TACKLES, NEW 6.0 MM CORRUGATED PLATE AND SCAFFOLDING MATERIALS ETC., FROM STORES TO SITE. 2. INSPECTION AND MARKING OF DAMAGED PORTIONS. 3. ERECTION OF SCAFFOLDING WHEREVER REQUIRED AND REMOVAL AND TRANSPORT TO STORES AFTER COMPLETION OF WORK. 4. DISMANTLING OF HONEY COMB PIECE BY CUTTING CAREFULLY AND GRINDING THE BURRS AND SLAG ON THE DUCT FLANGES. 5. KEEPING THE NEW HONEY COMB IN POSITION PROPERLY WELDING FULLY TO THE FLUE GAS DUCT.	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)
6	REPLACEMENT OF FLUE GAS DUCTS BRACINGS 80 NB SIZE. 1. INSPECTION AND MARKING OF ERODED CUTTING OF THE DAMAGED PIPES OR A PORTION OF DUCT & TAKING OUT THE DAMAGED PIECES FROM THE DUCT AND KEEPING AT THE SPECIFIED PLACE. CUTTING OF THE NEW PIPES OR 6 MM PLATE TO THE REQUIRED SIZE, TAKING THE SAME INSIDE THE DUCTS, LOCATING THE POSITION AND FULL WELDING OF THE PIPE / PLATE TO THE DUCT AT BOTH ENDS FIRMLY, FACING THE FLATS TO THE FLUE GAS FLOW DIRECTION.	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)

7	SEAM WELDING OF LEAKING PORTIONS IN DUCT 1. CHECK ANY LEAKS IN WELDING AND REWELD FIRMLY.	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STAND-ARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)
8	FABRICATION OF CORRUGATED M.S PLATE 6 MM THICKNESS. 1. ARRANGING NECESSARY T&P AT WORK AREA 2. TRANSPORTING OF M.S.CHEQ.PLATE6MM FROM STORES TO FABRICA-TION AREA. 3. MAKING ON PLATE AS PER EXISTING GUIDE VANE DIMENSION AND CUTTING INTO REQUIRED SIZE. 4. BENDING THE M.S. PLATE TO REQUIRED CURVATURE ON THE BEND-ING MACHINE. 5. TRANSPORTING THE BEND PLATES TO THE SPOT AT 16 MTRS ELEVA-TION OF BOILER. 6. CARRYING OUT THE MINOR ADJUSTMENTS TO SUIT THE FIELD CON-DITION BY SUITABLY HAMMERING AND HEATING IT. FABRICATION OF NEW GUIDE VANES AT APH FLUE GAS OUTLET DUCT BEND PORTION.	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STAND-ARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)



	FINAL INSPECTIONS & CHECKS: AS PER DIRECTIONS OF APGENCO/EIC.	VISUAL /DIMENSIONAL INSPECTION/SURFACE EXAMINATION NDT CHECKS /FUNCTIONAL CHECKS./	B	MEASUREMENTS /SURFACE EXAMINATION /DP/UT CHEKS /FUNCTIONAL CHECKS)	100%	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STANDARDS	AS PER APPROVED DRAWINGS/AS PER APPROVED /O&M MANUAL/PO ISO/MFD STANDARDS/ /MFD APPROVED ELETRODES/AWS WELDING/HT STAND-ARDS	M. MEASUREMENTS REGISTER RECORD	√	V (FUNCTIONAL CHECKS)

NOTE:

LEGEND: * RECORDS, INDENTIFIED WITH "TICK" (√) SHALL BE ESSENTIALLY INCLUDED BY SUPPLIER IN QA DOCUMENTATION. LEGEND TO BE USED: CLASS #: A = CRITICAL, B=MAJOR, C=MINOR; 'A' SHALL BE WITNESSED BY OWNER QUALITY DEPT., 'B' SHALL BE WITNESSED BY OWNER ERECTION / CONSTRUCTION DEPT. AND 'C' SHALL BE WITNESSED BY MAIN SUPPLIER (A & B CHECK SHALL BE OWNER CHECK STAGES).